



POWERING HYPERSCALE DATA CENTERS

A SUSTAINABLE 2N+1 POWER ARCHITECTURE FOR AI-DRIVEN WORKLOADS

OVERVIEW

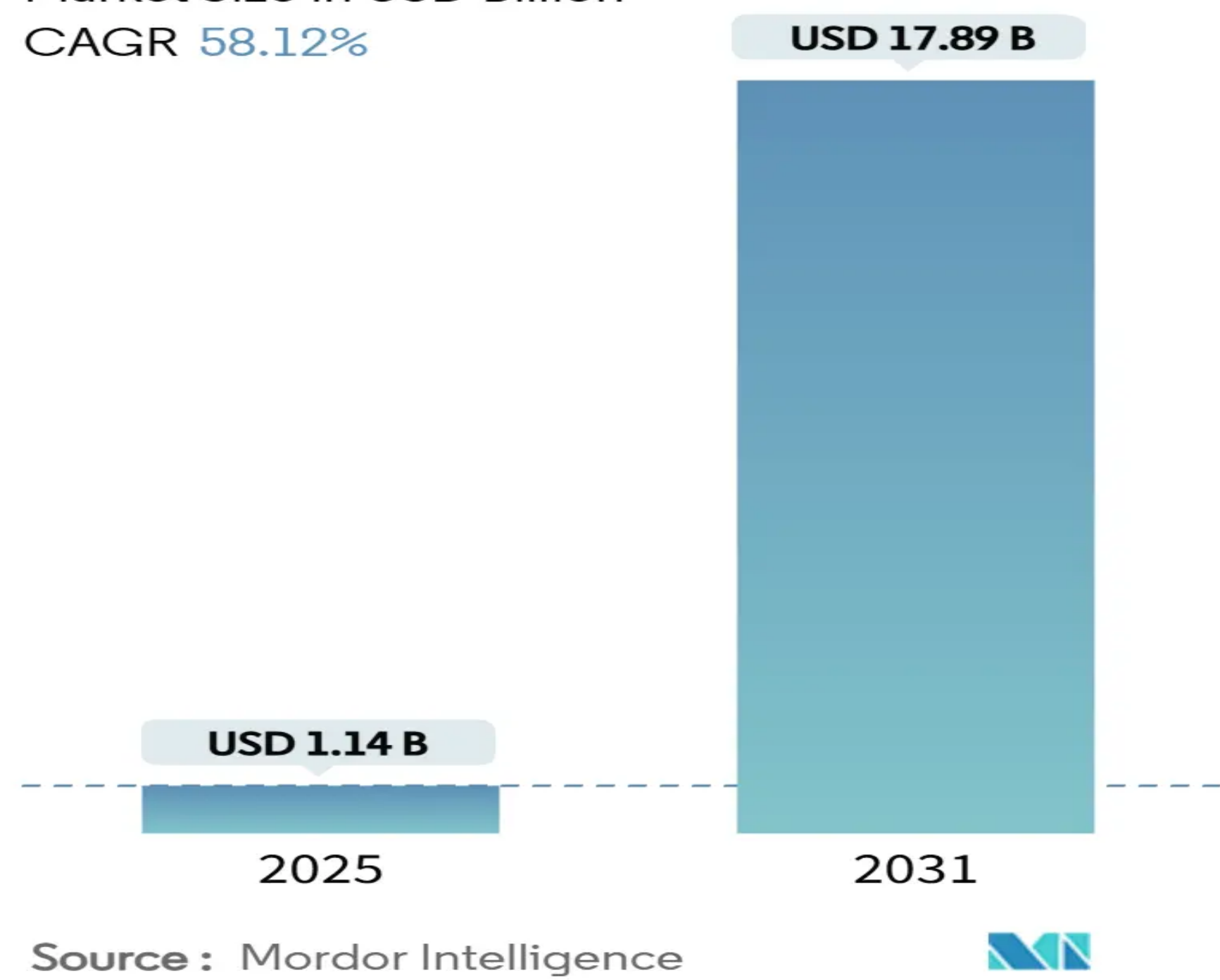
Data centers are projected to consume 8 – 12% of total U.S. electricity by 2030, up from 3 – 4% today. As AI workloads surge, maintaining continuous uptime is straining the grid. This project explores a hybrid 2N+1 power architecture integrating the utility grid, Small Modular Reactors (SMRs), and hydrogen standby generation to ensure resilient, sustainable power for hyperscale operations.

Context/Problem Statement

- Dublin, Ireland is the third-largest hyperscale data center market, behind Northern Virginia and Beijing
- Ireland's hyperscale data market is projected to reach \$17.89B by 2031 (58% CAGR)
- Dublin hosts 5% of hyperscalers worldwide, but faces a moratorium of grid connections above 50 MW
- Amazon, Microsoft, Meta and Google account for over 60% of hyperscalers and largest purchasers of renewable energy power purchase agreements (PPA)

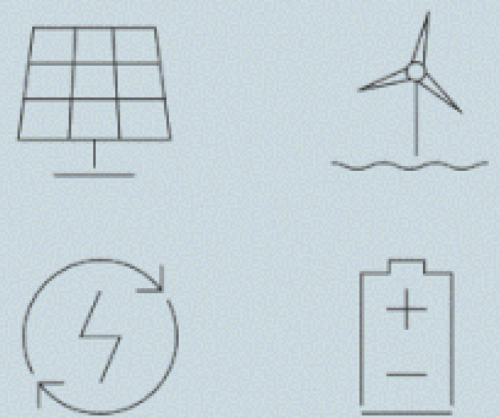
Ireland Hyperscale Data Center Market

Market Size in USD Billion
CAGR 58.12%

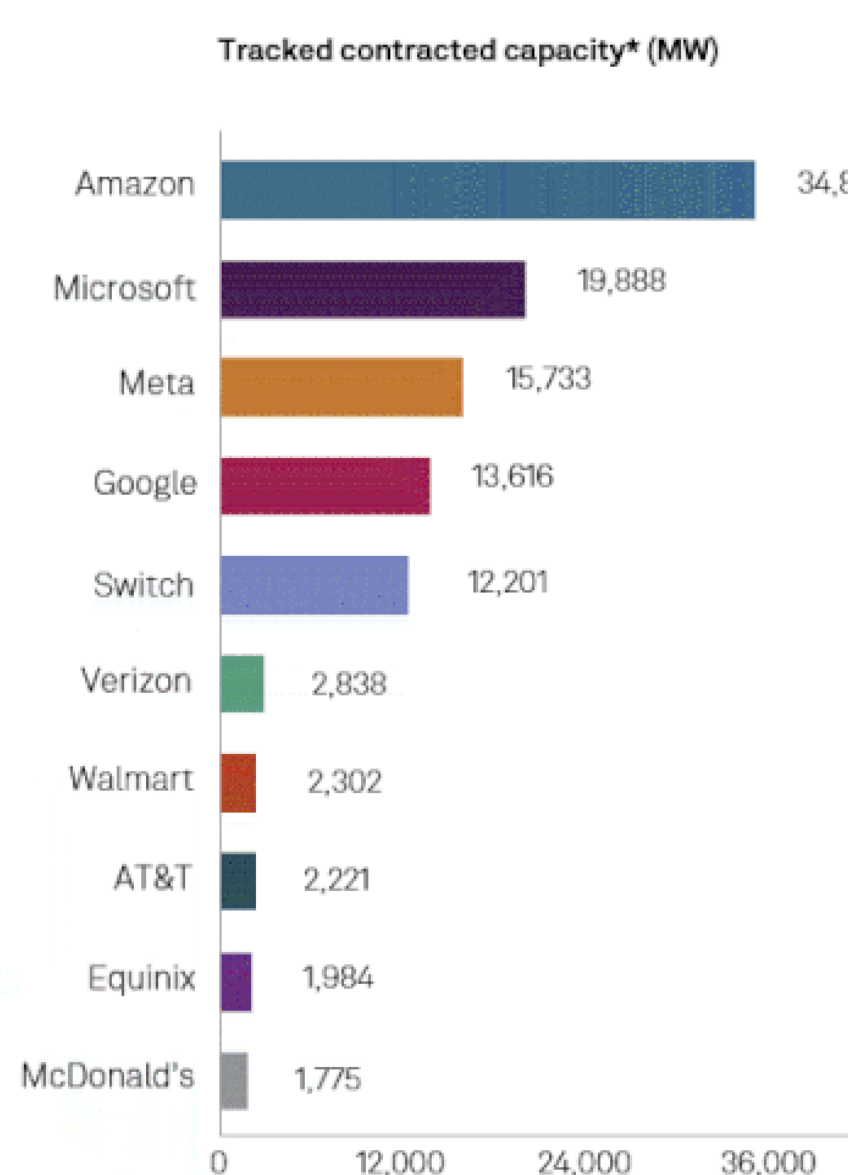


Clean energy corporate leaderboard

2025



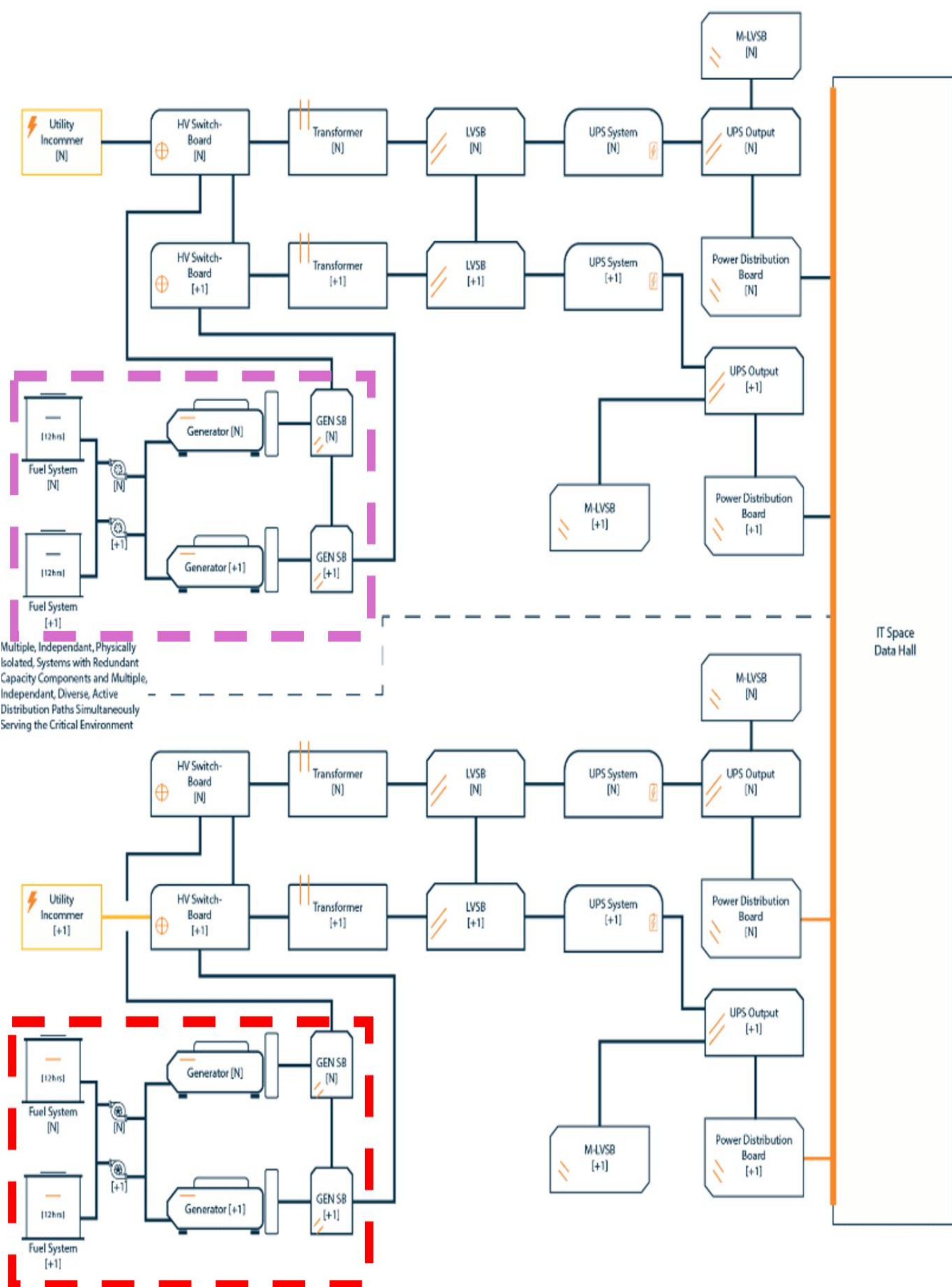
As of February 6, 2025.
* Reported and estimated. Cumulated carbon-free energy capacity contracted by US corporations only. All tracked deals.
Displayed years reflect annual S&P Global Commodity Insights tracking updates.
Total for Switch Inc. as of 2025 includes 20-year, nonbinding master power agreement with Oklo Inc. for 12 GW of nuclear capacity to be developed through 2044.
Analysis does not include most on-site corporate renewable capacity such as rooftop solar systems.
Sources: S&P Global Market Intelligence, public reports.
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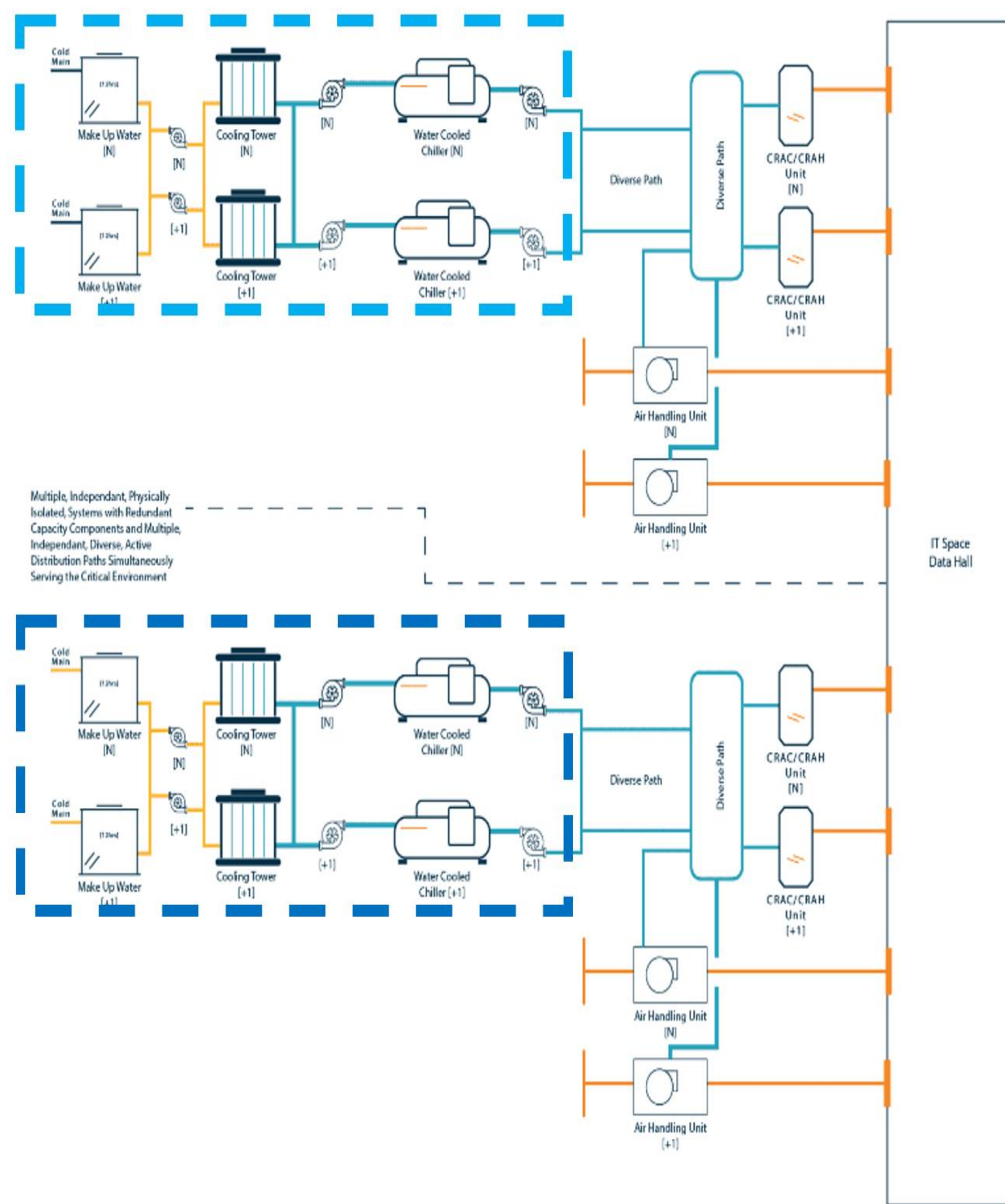
The 2N+1 Model

- 2N+1 redundancy provides maximum uptime and fault tolerance by supplying two independent power paths (2N) and one backup system (+1)
- Primary (N1): Utility grid, cheapest electricity source for baseline operations
- Secondary (N2): Generation IV Small Modular Reactor (Xe-100), provides 100% dispatchable, low-carbon baseload power
- Backup (+1): Hydrogen-fueled internal combustion engines or fuel cells, provide instant response to grid or SMR interruptions.
- Architecture enables Tier IV (fault-tolerant) performance with 99.999% uptime, ensuring uninterrupted AI compute, cooling, and network loads while supporting grid stability

Power System



Cooling System



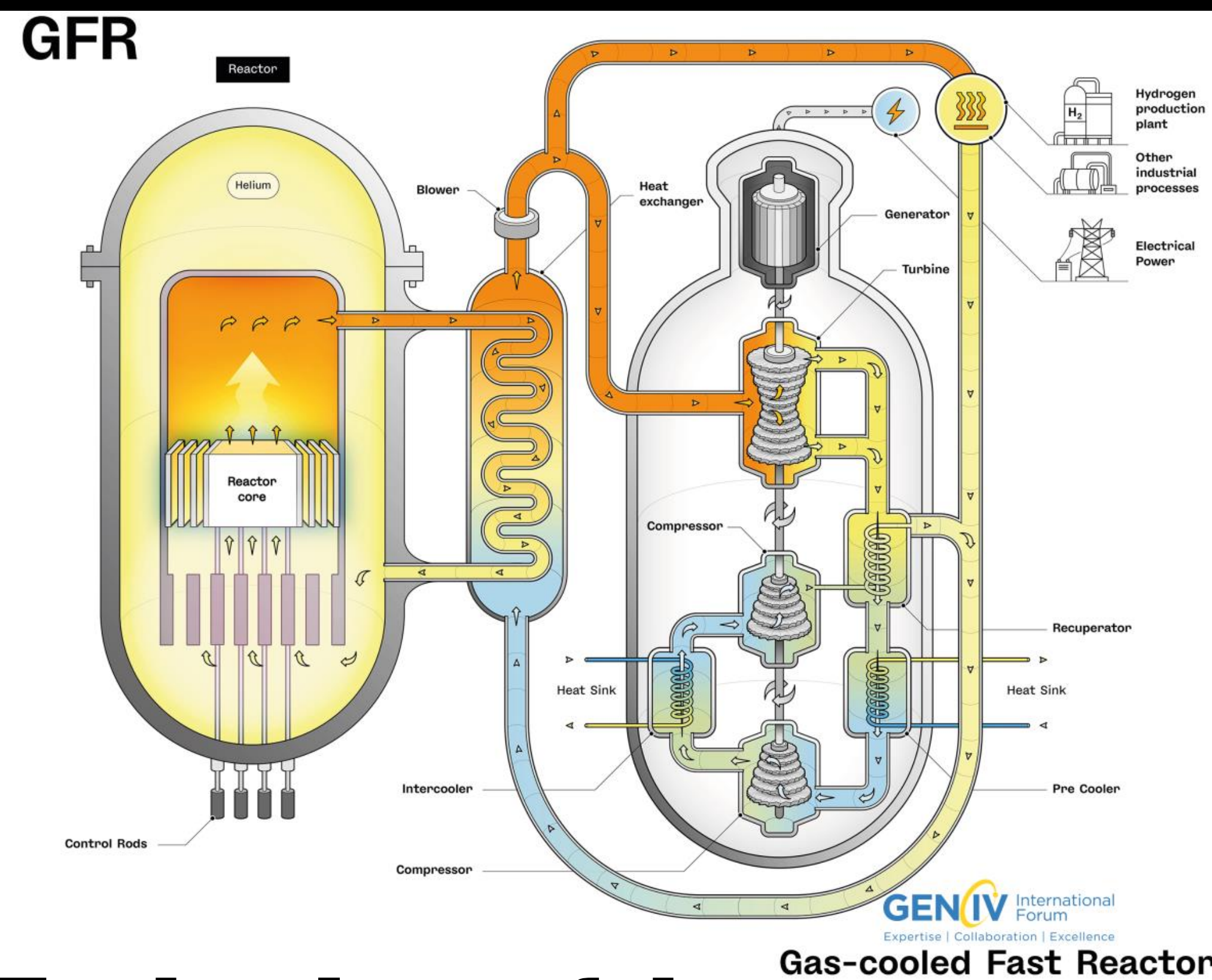
Financials

Operating SMRs is commercially de-risked by using the nuclear asset as a self-serving utility income stream. The core model relies on revenue stacking: securing fixed-price, long-term Power Purchase Agreements (**PPAs**) with credit-worthy hyperscalers (like AWS) alongside industrial revenue from clean heat/hydrogen to co-located partners (like Dow). This secured cash flow allows for the project to leverage low-cost debt financing (e.g., DOE loans), and therefore, vastly improves the project's **Internal Rate of Return (IRR)** over its operating lifespan.

X-energy Project Funding and Revenue Streams



Gas-Cooled Fast Reactor



Technology of the Small Modular Reactor

Xe-100 from X-energy

- Generation IV nuclear reactor (Gas cooled fast reactor)
- Utilizes TRISO fuel: Tri-structural ISotropic particle fuel
- TRISO fuel cannot melt in a commercial high-temp reactor (U.S. Dept. of Energy)
- Can be base load or load following
- 400-meter safety perimeter (as opposed to 10 miles)